Operations Research, Spring 2026 (114-2)
Final Project Proposal

Group 4

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1 Introduction

Fantasy Baseball is a strategic game where participants (managers) act as general managers to build a team that competes based on the real-life statistical performance of Major League Baseball (MLB) players. Among various league formats, the Head-to-Head (H2H) Points league is one of the most popular. In this format, players' real-life statistics (e.g., home runs, stolen bases, strikeouts) are translated into fantasy points based on a predetermined weighting system.

Before the MLB season begins, managers construct their rosters through a "Snake Draft." In a snake draft, managers take turns selecting players from a pool of available MLB players. The order reverses every round (e.g., in a 12-team league, the manager with the 1st pick in Round 1 gets the 24th pick in Round 2). The primary goal during the draft is to construct a roster that maximizes the total expected fantasy points for the season.

However, formulating a draft strategy is highly challenging due to three factors. First, a valid roster must satisfy strict positional constraints (e.g., exactly 1 Catcher, 1 First Baseman, 3 Outfielders, etc.). Second, many players have multi-position eligibility, meaning they can be flexibly slotted into different positions. Third, the draft is a highly dynamic competitive environment. A manager cannot simply select the players with the highest projected points because opponents will also draft players, causing the available player pool to shrink continually. The market value or expected availability of a player is typically measured by their Average Draft Position (ADP), which represents the consensus overall pick number at which a player is drafted across thousands of mock drafts.

Traditional drafting strategies often rely on naive heuristics, such as selecting the "Best Player Available" based on raw projected points, which often leads to severe positional imbalances (e.g., drafting too many First Basemen while lacking a valid Shortstop). To address this, we propose an Operations Research approach. In this project, we model the Fantasy Baseball Snake Draft as an Integer Programming (IP) problem. By incorporating ADP data to simulate the dynamic availability of players and utilizing projected points as the objective, we aim to develop a decision-support model that recommends the optimal sequence of draft picks to maximize a team's total projected points while strictly adhering to roster constraints.

In particular, we use this optimization framework to quantify the structural advantage of different draft positions in a snake draft by comparing the total projected fantasy points attainable from each draft slot.

2 Problem description

We consider the decision-making process of a single fantasy manager participating in a snake draft. The manager possesses a set of specific overall draft picks assigned by the snake draft format. The objective is to determine exactly which player to select at each of the manager's assigned draft picks to maximize the sum of the projected points of the final roster.

There are several physical and competitive constraints that must be satisfied. Firstly, the final roster must consist of a specific number of players assigned to required physical positions: Catcher (C), First Baseman (1B), Second Baseman (2B), Third Baseman (3B), Shortstop (SS), Outfielders (OF), Utility (Util), Starting Pitchers (SP), and Relief Pitchers (RP).

Secondly, player eligibility must be respected. A drafted player can only be assigned to a position for which they are eligible in real life. If a player possesses multi-position eligibility (e.g., eligible for both 2B and SS), the model must assign them to exactly one of those positions to fulfill the roster requirements.

Thirdly, and most importantly, we must account for the dynamic availability of players using the Average Draft Position (ADP). In reality, a manager cannot draft a highly-ranked player in the later rounds because opponents would have already selected them. To model this, we introduce an availability constraint: if a manager intends to draft a specific player using their k-th assigned pick (which corresponds to a specific overall draft slot), the player's ADP must not be significantly lower than that overall draft slot. If the player's ADP indicates they will likely be drafted much earlier, the player is considered unavailable for that pick and all subsequent picks.

In this project, we apply the model in an offline simulation setting focused on draft-position analysis. For each historical season and each possible draft position, we treat the manager's set of picks as given, solve the ILP to obtain an optimal sequence of draft picks, and use the total projected points of the resulting roster as the measure of performance for that draft position.

3 Mathematical model

In this section, we formulate an Integer Linear Program (ILP) to describe the draft optimization problem for a manager from a given state in the draft. To simplify the analysis, we ignore daily roster moves and bench management: we assume that every drafted player is always started and that their full-season projected points already capture their contribution to the manager's score.

Let $I = \{1, ..., n\}$ be the set of available players in the player pool, P be the set of required roster positions (e.g., $P = \{C, 1B, 2B, 3B, SS, OF, Util, SP, RP\}$). Let K be the set of the manager's remaining draft picks in the snake draft; and for each remaining pick $k \in K$ in a snake draft with M managers participated, where the decision maker starts at initial selecting position j ($1 \leq j \leq M$), the slot number for any given integral round $1, ..., |P|$ depending on the sequence direction is shown down below:

$$\begin{cases} \text{Forward Rounds } (r \text{ is odd}): & k = (r-1)M + j \\ \text{Reverse Rounds } (r \text{ is even}): & k = rM - j + 1. \end{cases}$$

Then by consolidating the two cases, we can get the following equation corresponding to round r, which could be used in later calculation:

$$k = (r-1)M + \frac{M+1}{2} + (-1)^r \left(\frac{M+1}{2} - j \right) \quad \forall k \in K, r = 1, ..., |P|. \tag{1}$$

Parameters:

- V_i: The projected fantasy points for player i for the entire season.
- $E_{ip} \in \{0, 1\}$: 1 if player i is eligible to play position p, and 0 otherwise.

- R_p: The required number of players to fill position p on the roster.
- S_k: overall draft slot number corresponding to pick ($k \in K$).
- A_i: The Average Draft Position (ADP) of player i.
- δ: A non-negative buffer parameter representing the volatility or standard deviation of the ADP (to allow drafting players slightly past their ADP).

Decision Variables: We define two sets of binary decision variables to handle pick assignment and positional assignment:

- $z_{ik} \in \{0, 1\}$: 1 if player i is drafted by the manager using their k-th pick, and 0 otherwise.
- $x_{ip} \in \{0, 1\}$: 1 if player i is assigned to roster position p, and 0 otherwise.
- $y_i \in \{0, 1\}$: 1 if player i is drafted onto the manager's roster at any pick, and 0 otherwise.

Objective Function: The total projected points of a drafted roster the manager seeks to maximize is then:

$$\max \quad \sum_{i \in I} V_i y_i \tag{2}$$

Constraints:

$$\sum_{i \in I} z_{ik} = 1 \quad \forall k \in K \tag{3}$$

$$\sum_{k \in K} z_{ik} = y_i \quad \forall i \in I \tag{4}$$

$$\sum_{p \in P} x_{ip} = y_i \quad \forall i \in I \tag{5}$$

$$x_{ip} \leq E_{ip} \quad \forall i \in I, p \in P \tag{6}$$

$$\sum_{i \in I} x_{ip} = R_p \quad \forall p \in P \tag{7}$$

$$z_{ik} = 0 \quad \forall i \in I, k \in K \text{ such that } A_i + \delta < S_k \tag{8}$$

$$z_{ik}, x_{ip}, y_i \in \{0, 1\} \quad \forall i \in I, k \in K, p \in P \tag{9}$$

Constraint (3) ensures that exactly one player is selected at each of the manager's draft picks. Constraint (4) links variables z_{ik} and y_i, ensuring that a player is considered drafted on the roster ($y_i = 1$) if and only if they are selected at one of the k picks. Note that this also implicitly prevents a player from being drafted more than once by the manager since $y_i \leq 1$ due to the binary constraint. Constraint (5) dictates that if a player is drafted, they must be assigned to exactly one position on the roster. Constraint (6) guarantees that a player can only be assigned to a position for which they are eligible. Constraint (7) enforces the exact physical roster requirements for each position p. Constraint (8) is the critical availability constraint. It forcefully prevents the model from assigning player i to pick k if the overall slot S_k is significantly later than the player's expected market depletion point $(A_i+\delta)$. Constraint (9) defines the binary nature of the decision variables.

4 Expected results

To conduct this study, we will use completed historical MLB seasons (e.g., 2021–2024) for which reliable pre-season projections and ADP data are available. For each season and each draft slot $j = 1, \dots, M$, we fix the manager's draft position, solve our ILP to construct an optimal roster under the ADP-based availability constraints, and record the total projected fantasy points $\sum_{i \in I} V_i y_i$ of the resulting roster as the performance metric for that slot. Aggregating and averaging these values across multiple seasons allows us to quantify the structural advantage of early, middle, and late draft positions under our model. Since our main goal is to analyze the structural advantage of different draft orders rather than to forecast a specific future season, running our model on historical data using projected points as the primary evaluation metric is sufficient.

We plan to implement the proposed ILP model in Python using the PuLP library or Gurobi Optimizer. To evaluate the performance of our model, we will simulate a 12-team snake draft environment. We will compare our ILP strategy against conventional heuristic algorithms, such as the "Greedy / Best Player Available" approach (always selecting the player with the highest V_i without considering ADP) and a "Static IP" approach (solving the roster without dynamic ADP constraints). We expect our proposed model to significantly outperform the naive heuristics in terms of total projected points while strictly adhering to positional constraints.

Furthermore, we will conduct a sensitivity analysis based on OR theories covered in class. By relaxing certain constraints, we can observe the Shadow Price (Dual variables) associated with each positional constraint (7). We anticipate that scarce positions (e.g., Shortstop or Catcher) will exhibit much higher shadow prices compared to abundant positions (e.g., Outfield), quantitatively proving the concept of "positional scarcity" in fantasy sports. We will also test the robustness of our model by varying the ADP buffer parameter δ to see how risk-averse or risk-seeking behaviors affect the optimal draft path.